

MDS-UPDRS Finger Tapping Score

The data provided below consists of extracted features (tap count, periods, and amplitudes) derived from a subject performing the MDS-UPDRS Item 3.4 Finger Tapping test. The subject was instructed to tap the index finger on the thumb 10 times as quickly and as largely as possible.

The objective is to score the subject according to MDS-UPDRS item 3.4: FINGER TAPPING based on these 10 consecutive finger taps. The testing instructions and scoring criteria for MDS-UPDRS item 3.4: FINGER TAPPING are as follows:

Instructions to the examiner:

Each hand is tested separately. Demonstrate the task, but do not continue to perform the task while the patient is being tested.

Instruct the patient to tap the index finger on the thumb 10 times as quickly AND as large as possible. Rate each side separately, evaluating speed, amplitude, hesitations, halts, and decrementing amplitude.

Scoring Criteria:

0 (Normal): No problems

1 (Slight): Any of the following:

- a) The regular rhythm is broken with one or two interruptions or hesitations of the tapping movement
- b) Slight slowing
- c) The amplitude decrements near the end of the 10 taps

2 (Mild): Any of the following:

- a) 3 to 5 interruptions during tapping
- b) Mild slowing
- c) The amplitude decrements midway through the 10-tap sequence

3 (Moderate): Any of the following:

- a) More than 5 interruptions during tapping or at least one longer arrest (freeze) in ongoing movement
- b) Moderate slowing
- c) The amplitude decrements starting after the 1st tap

4 (Severe): Cannot or can only barely perform the task due to slowing, interruptions, or decrements

Below are the extracted features including tap count, periods, and amplitudes. Please analyze this information and provide your score on the last line.

EXTRACTED FEATURES

Number of taps: 11

Tap periods (frames): [150, 103, 96, 113, 101, 108, 77, 116, 101, 96]

Tap amplitudes (pixels): [144.80331487918363, 128.37834708392222, 126.62543188475212, 110.01818031580054, 137.82960494755835, 134.23859355639868, 128.3627671873741, 124.77579893553076, 119.40686747419514, 127.45587471748802, 113.63538181394033]

Gemini 3 Pro

Based on the extracted features provided, here is the analysis and scoring for MDS-UPDRS Item 3.4: FINGER TAPPING.

Analysis

1. Speed and Rhythm (Tap Periods):

- **Data:** [150, 103, 96, 113, 101, 108, 77, 116, 101, 96]
- **Observation:** The tap periods are relatively consistent, averaging around 100-110 frames (excluding the initial start-up period of 150). There is one faster tap (77) and one slightly slower tap (116), but there are no significant pauses, arrests, or "freezes" (which would appear as very large period values). The rhythm is not significantly broken or interrupted.

2. Amplitude (Tap Amplitudes):

- **Data:** [144.8, 128.4, 126.6, 110.0, 137.8, 134.2, 128.4, 124.8, 119.4, 127.5, 113.6]
- **Observation:**
 - **Sequence 1 (Taps 1-4):** The amplitude drops immediately after the first tap (144.8 $\rightarrow$ 128.4) and continues to decrement to Tap 4 (110.0).
 - **Recovery:** There is a significant recovery in amplitude at Tap 5 (137.8), returning close to the initial maximum.
 - **Sequence 2 (Taps 5-9):** From Tap 5 onwards, there is a consistent, gradual decrement in amplitude: 137.8 $\rightarrow$ 134.2 $\rightarrow$ 128.4 $\rightarrow$ 124.8 $\rightarrow$ 119.4.

- **Recovery:** There is another slight recovery at Tap 10.

Scoring Rationale

- **Score 0 (Normal):** Incorrect. There are clear decrements in amplitude.
- **Score 1 (Slight):** Requires amplitude decrement "near the end". The data shows decrementing trends starting early (Tap 1) and again midway (Tap 5), which is more significant than just "near the end".
- **Score 3 (Moderate):** Requires amplitude decrement "starting after the 1st tap". While the amplitude does drop after the 1st tap, the subject demonstrates the ability to recover amplitude significantly (at Tap 5), which suggests the impairment is not "Moderate" or "Severe" in nature (where amplitude usually crushes down and stays small).
- **Score 2 (Mild):** The criteria states: "The amplitude decrements midway through the 10-tap sequence". The sequence from Tap 5 to Tap 9 represents a clear, continuous decrement starting midway through the test. This fits the "Mild" criteria best, as the decrement is obvious and persistent in the second half, but the amplitude remains functional (pixels > 100) and recovers, distinguishing it from a more severe score.

Score: 2
